

Problem

Find the phasors corresponding to the following signals:

(a) $v(t) = 21 \cos(4t - 15^\circ) \text{ V}$

(b) $i(t) = -8 \sin(10t + 70^\circ) \text{ mA}$

(c) $v(t) = 120 \sin(10t - 50^\circ) \text{ V}$

(d) $i(t) = -60 \cos(30t + 10^\circ) \text{ mA}$

Step-by-step solution

Step 1 of 4

(a)

Consider the following signal representation:

$$v(t) = 21 \cos(4t - 15^\circ) \text{ V}$$

Here,

The amplitude, $V_m = 21 \text{ V}$,

The angular frequency, $\omega = 4 \text{ rad/s}$,

The phase angle, $\phi = -15^\circ$.

The phasor is a complex number that represents the amplitude and phase of a sinusoid.

The following is the phasor for the signal.

$$v(t) = 21 \angle -15^\circ \text{ V}$$

Thus, the phasor for the signal, $v(t)$ is $21 \angle -15^\circ \text{ V}$.

Step 2 of 4

(b)

Consider the following signal representation:

$$i(t) = -8 \sin(10t + 70^\circ) \text{ mA}$$

Represent the signal in cosine form for easy understanding.

$$\begin{aligned} i(t) &= 8 \sin(180^\circ + 10t + 70^\circ) \text{ mA} && (\text{since, } \sin(\pi + \theta) = -\sin \theta) \\ &= 8 \cos(90^\circ - 180^\circ - 10t - 70^\circ) \text{ mA} && \left(\text{since, } \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta \right) \\ &= 8 \cos(-10t - 160^\circ) \text{ mA} \\ &= 8 \cos(10t + 160^\circ) \text{ mA} && (\text{since, } \cos(-\theta) = \cos \theta) \end{aligned}$$

Here,

The amplitude, $I_m = 8 \text{ mA}$,

The angular frequency, $\omega = 10 \text{ rad/s}$,

The phase angle, $\phi = 160^\circ$.

The phasor is a complex number that represents the amplitude and phase of a sinusoid.

The following is the phasor for the signal.

$$i(t) = 8 \angle 160^\circ \text{ mA}$$

Thus, the phasor for the signal, $i(t)$ is $\boxed{8 \angle 160^\circ \text{ mA}}$.

Step 3 of 4

(c)

Consider the following signal representation:

$$v(t) = 120 \sin(10t - 50^\circ) \text{ V}$$

Represent the signal in cosine form for easy understanding.

$$\begin{aligned} v(t) &= 120 \cos(90^\circ - 10t + 50^\circ) \text{ V} && \left(\text{since, } \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta \right) \\ &= 120 \cos(-10t + 140^\circ) \text{ V} \\ &= 120 \cos(10t - 140^\circ) \text{ V} && (\text{since, } \cos(-\theta) = \cos \theta) \end{aligned}$$

Here,

The amplitude, $V_m = 120 \text{ V}$,

The angular frequency, $\omega = 10 \text{ rad/s}$,

The phase angle, $\phi = -140^\circ$.

The phasor is a complex number that represents the amplitude and phase of a sinusoid.

The following is the phasor for the signal.

$$v(t) = 120 \angle -140^\circ \text{ V}$$

Thus, the phasor for the signal, $v(t)$ is $\boxed{120 \angle -140^\circ \text{ V}}$.

Step 4 of 4

(d)

Consider the following signal representation:

$$i(t) = -60 \cos(30t + 10^\circ) \text{ mA}$$

Represent the signal in cosine form for easy understanding.

$$\begin{aligned} i(t) &= 60 \cos(180^\circ - 30t - 10^\circ) \text{ mA} && (\text{since, } \cos(\pi - \theta) = -\cos \theta) \\ &= 60 \cos(-30t + 170^\circ) \text{ mA} \\ &= 60 \cos(30t - 170^\circ) \text{ mA} && (\text{since, } \cos(-\theta) = \cos \theta) \end{aligned}$$

Here,

The amplitude, $I_m = 60 \text{ mA}$,

The angular frequency, $\omega = 30 \text{ rad/s}$,

The phase angle, $\phi = -170^\circ$.

The phasor is a complex number that represents the amplitude and phase of a sinusoid.

The following is the phasor for the signal.

$$i(t) = 60 \angle -170^\circ \text{ mA}$$

Thus, the phasor for the signal, $i(t)$ is $60 \angle -170^\circ \text{ mA}$.